

The Bibliometric Analysis of EEGLAB software in the Web of Science indexed Articles



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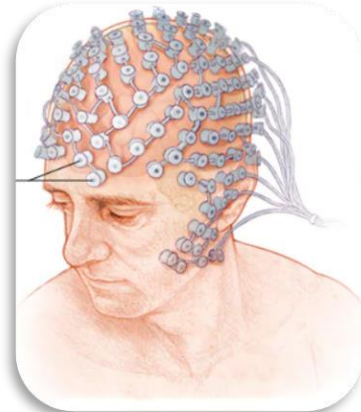


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Introduction

EEGLAB is one of the most famous software for processing, analyzing, and researching experiments that have Electroencephalography (EEG) datasets. Due to the numerous and famous add-ins along with global, widespread communications and online free training, its popularity increased every year. [1] In this study, we present the bibliometric analysis with some text mining methods for aggregated abstracts by using the bibliometrix R package for all available ISI research articles that have been cited the EEGLAB. [2]

Highlights

- Summarizes 12,700 ISI-indexed articles about EEGLAB.
- Clustered Collaboration Network University into 6 segments.
- Presented the trend topics plot for keyword plus.
- Presented Co-Citation Network of authors for all and core sources.
- Have a big appendix for further analysis and reproducible results.

Results

The time span of these articles is from 2004 to 2023 with 12,700 documents in 1,125 sources (journals, books, etc.), 29,125 authors, 19,062 author's keywords, 13,707 keywords PLUS, 279,617 references. The annual growth rate is 28.12 %, international Co-authorship is 37.27 % and Co-authors per document is s 4.89 and the average citations per document is 22.51. The most relevant sources are Neuroimage, Frontiers in Human Neurosciences, Scientific Reports, Psychophysiology, and PLOS One with 780, 526, 446,425, and 371 articles, respectively. The most cited countries are the USA, Germany, and the United Kingdom with 93,093, 32,621, and 20,748 total citations, respectively. The ERPLAB, ADJUST, and ICLabel add-ins have the local to global citation ratios equal to 85.4%, 65.1%, and 78.2% respectively. Other bibliometric analyses such as co-occurrence networks and thematic maps of abstracts, titles, and keywords are estimated and presented.

Methods

To address this phenomenon from a bibliographic perspective, we found 20,464 citations in Google Scholar for the main EEGLAB reference [1] since 8/27/2023. Then, only the Web of Science (WOS) articles were 12,700 that they were extracted. The results were analyzed with Bibliometrix package from CRAN R software.

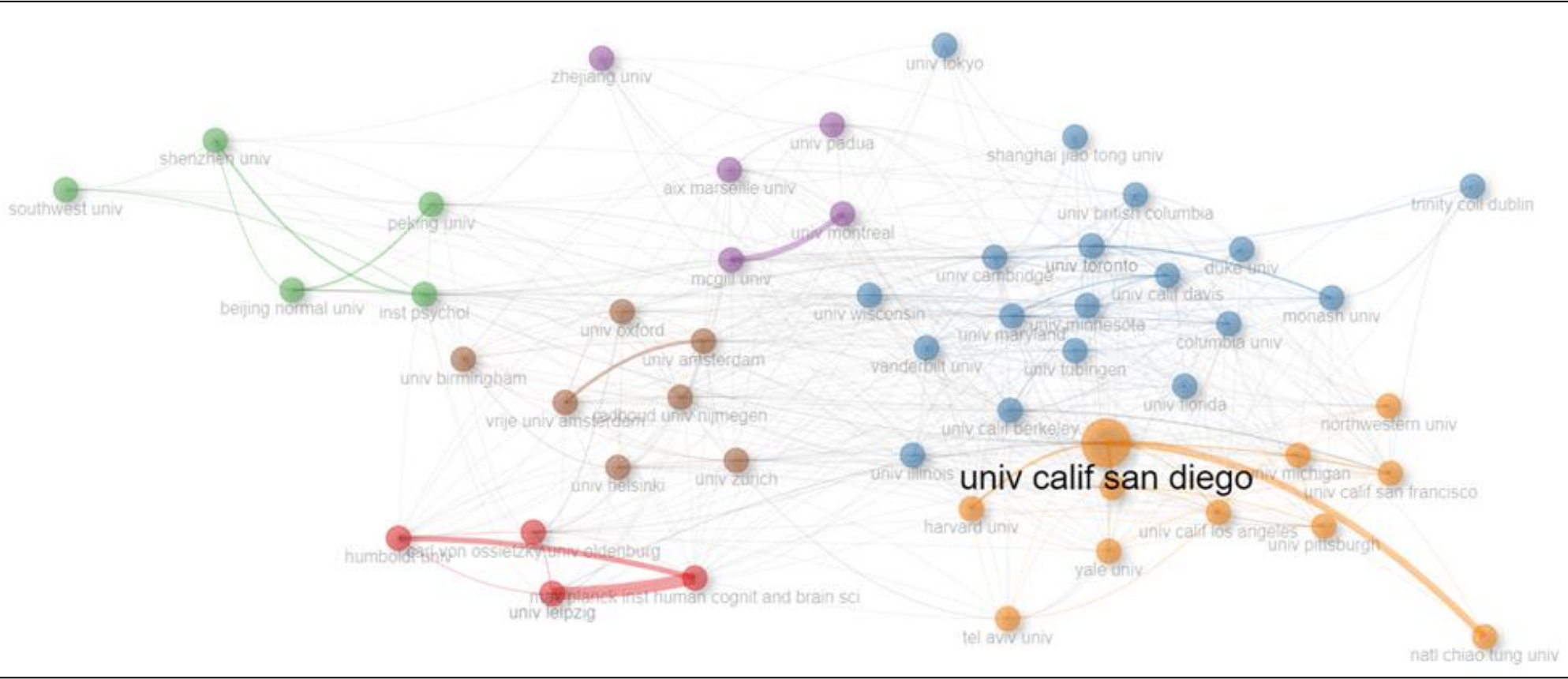


Figure 1 – Collaboration Network University (6 Clusters)

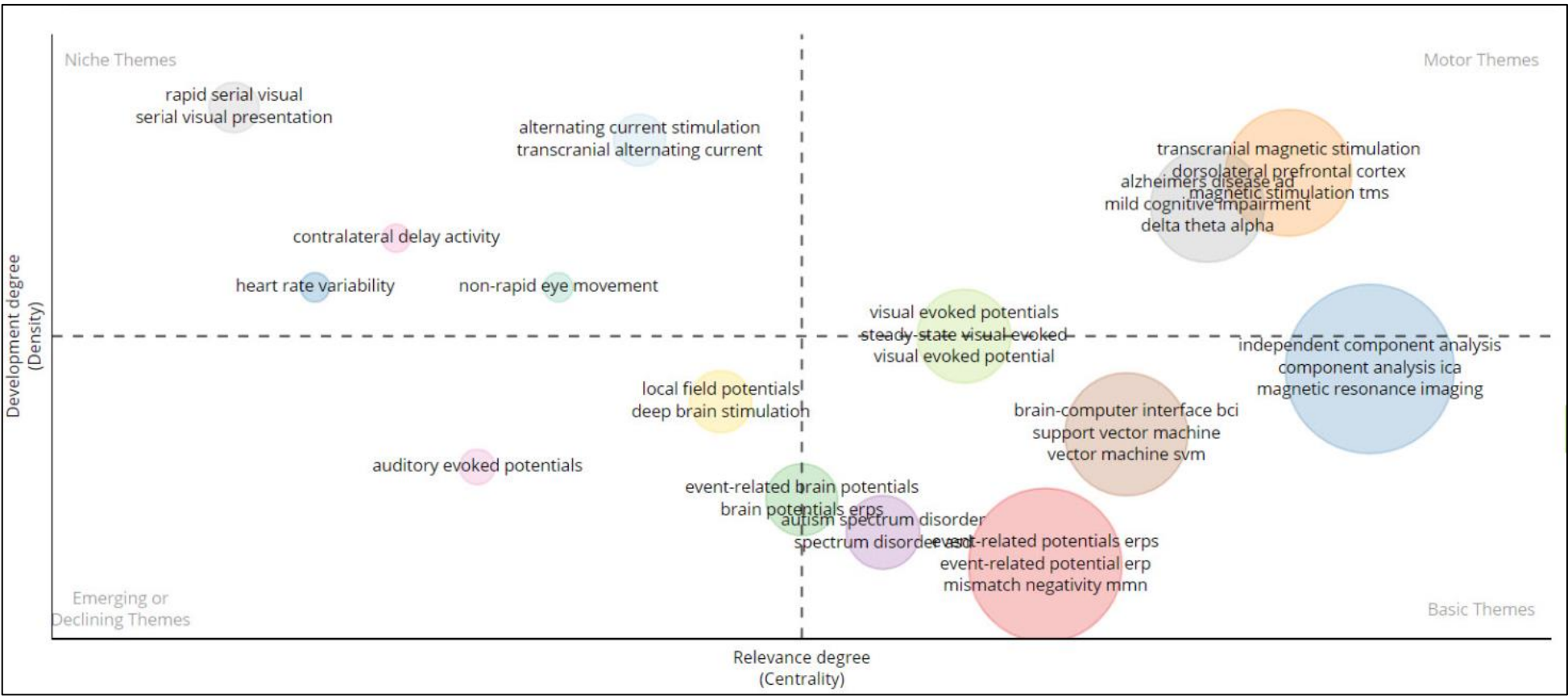


Figure 2 - Thematic Map Trigram Word in Abstract

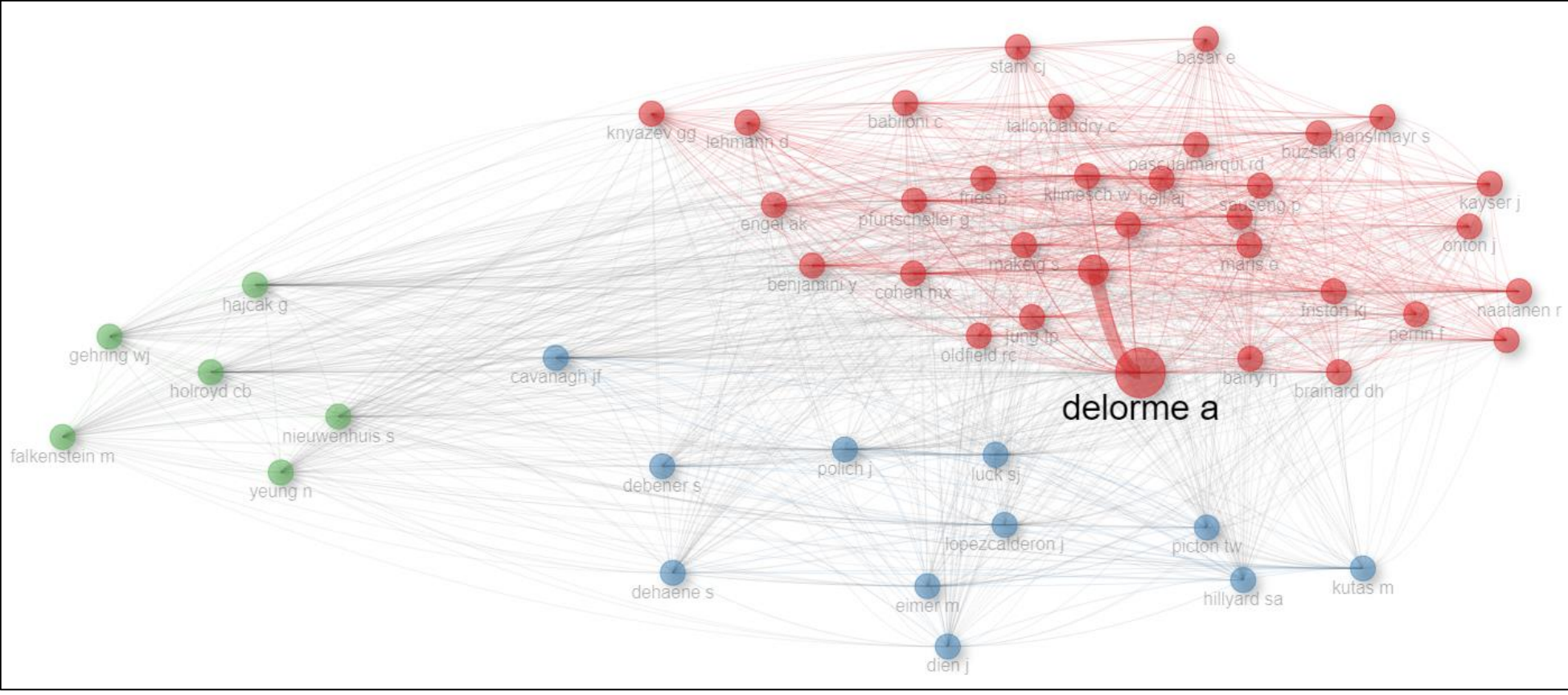


Figure 3 – Co-Citation Network of Authors (3 Clusters)

Table 1 - The Most Local Cited Documents						
Row	Ref	Description	Publication Year	Citations		
				Local	Global	Ratio
1	[3]	ERPLAB app	2014	1215	1422	85.4
2	[49]	ADJUST app	2011	529	812	65.1
3	[43]	ICA – Artifacts Detection	2007	524	1087	48.2
4	[44]	ERP	2004	487	1014	48.0
5	[4]	ICLabel app	2019	392	501	78.2
6	[50]	ICA and BSS	2012	311	536	58.0
7	[51]	multiple comparison correction	2011	296	716	41.3
8	[48]	Coupling EEG/fMRI	2005	233	837	27.8
9	[52]	log spectral ICA	2005	229	590	38.8
10	[53]	ERP - Theta band	2012	172	428	40.2

Conclusion

EEGLAB is among the most cited MATLAB toolboxes in computational neuroscience. Many developed and developing countries use it in their research publications. Many global and famous universities and research institutes published research with EEGLAB in the USA, Europe, Canada, Japan, Australia and Russia. But it is not limited to developed countries, and many developing countries like China, India, Taiwan, Turkey, Iran, Saudi Arabia, Bangladesh, Brazil, Cuba, Argentina, Colombia, and many others use it in their scientific experiments and publications.

References

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Figure 4 – Most Relevant Trigram words in Title

